

Project Case	
COSC6050003 COSC6067052 Programming for Business	
Computer Science	0272-COSC6050-HW02-00
<i>Valid on Odd Semester Year 2026/2027</i>	Revision 00

1. Mahasiswa tidak diperkenankan untuk:

Students are prohibited from:

- Berdiskusi dan/atau bekerja sama dengan mahasiswa lainnya,
Discussing and/or cooperating with other students,
- Melihat sebagian atau seluruh jawaban mahasiswa lainnya,
Seeing a part or the whole answer from other students,
- Membuka dan/atau menyalin jawaban dari buku, catatan, video, dan jenis referensi lainnya,
Open and/or copy answer from books, notes, videos, and other references,
- Membuka dan/atau menyalin jawaban dari internet,
Open and/or copy answer from the internet,
- Mengumpulkan jawaban yang tidak sesuai dengan tema soal,
Submitting an answer with a different theme from the given case,
- Melakukan tindakan yang menyebabkan jawaban dicontek oleh orang lain atau kelompok lain, baik disengaja maupun tidak disengaja,
Doing action that could result the answer being copied by someone or other groups, intentionally or unintentionally,
- Melakukan tindakan kecurangan lainnya.
Committing other dishonest actions.

2. Jika mahasiswa dan/atau terbukti melakukan tindakan seperti yang dicantumkan pada butir ke-1, maka nilai mahasiswa dan/atau kelompok yang melakukan kecurangan, baik menyontek atau dicontek, akan dinolkan sesuai dengan peraturan yang berlaku.

If it has been proven that a student and/or group has committed dishonest actions outlined in point 1 above, the students and/or groups related to the incident, regardless of which one copies or has their answer copied, will be issued a score of zero according to the regulation.

3. Jawaban yang dapat diterima dan dinilai adalah jawaban yang dikumpulkan sebelum batas waktu yang telah ditentukan.

The answer must be submitted before the designated deadline to be accepted and graded,

4. Jawaban akan dinilai berdasarkan teknik yang diajarkan dalam praktikum dengan menggunakan software yang telah ditentukan.

The scoring will be based on the materials taught during the practicum classes using the designated software. Using different software than requested may result in your answer not being graded.

5. Jika Anda tidak membaca peraturan ini, maka Anda dianggap sudah membaca dan menyetujuinya.

By taking this exam, you agree to these regulations, regardless of whether you have read it or not.

6. Persentase penilaian untuk matakuliah ini adalah sebagai berikut:

The score will be distributed as follows:

Tugas Mandiri <i>Assignment</i>	Proyek <i>Project</i>	UAP <i>Final Exam</i>
40%	60%	-

7. Perangkat lunak yang digunakan pada matakuliah ini adalah sebagai berikut:

This course uses the following software:

Software <i>Software</i>
XAMPP 8.2.12 Visual Studio Code

8. Ekstensi file yang harus dikumpulkan untuk matakuliah ini adalah sebagai berikut:

Your answers must be in the following file extensions:

Tugas Mandiri <i>Assignment</i>	Proyek <i>Project</i>	UAP <i>Final Exam</i>
HTML, CSS, JS	PHP	-

Soal*Case*

CineRent

CineRent is an online DVD/Blu-ray rental platform where members can browse available films and place rental requests. The website will be managed by the admin, allowing them to manage users, films, genres, and rental transactions. As a web developer, you are asked to create a website using PHP for this platform. This means you are expected to use **HTTP requests** to handle all **Create, Read, Update, and Delete (CRUD) operations**.

There are three (3) types of user roles on this website: **Admin**, **Member**, and **Guest** (non-logged-in user). Minimum pages you need to create for each role are:

- **Admin**
 - Home (Dashboard)
 - Manage Films
 - Add and Edit Film
 - Manage Genres
 - Manage Users
 - Manage Rentals
- **Member**
 - Home
 - Film Catalog
 - Film Detail
 - Cart
 - My Rentals
 - Profile
- **Guest**
 - Home
 - Login
 - Register

Each page shows the name of the **Member or Admin**. If the user has not logged in (**Guest**), then they can only access Guest pages. You need to build this website based on the following criteria:

1. Entity Relationship Diagram

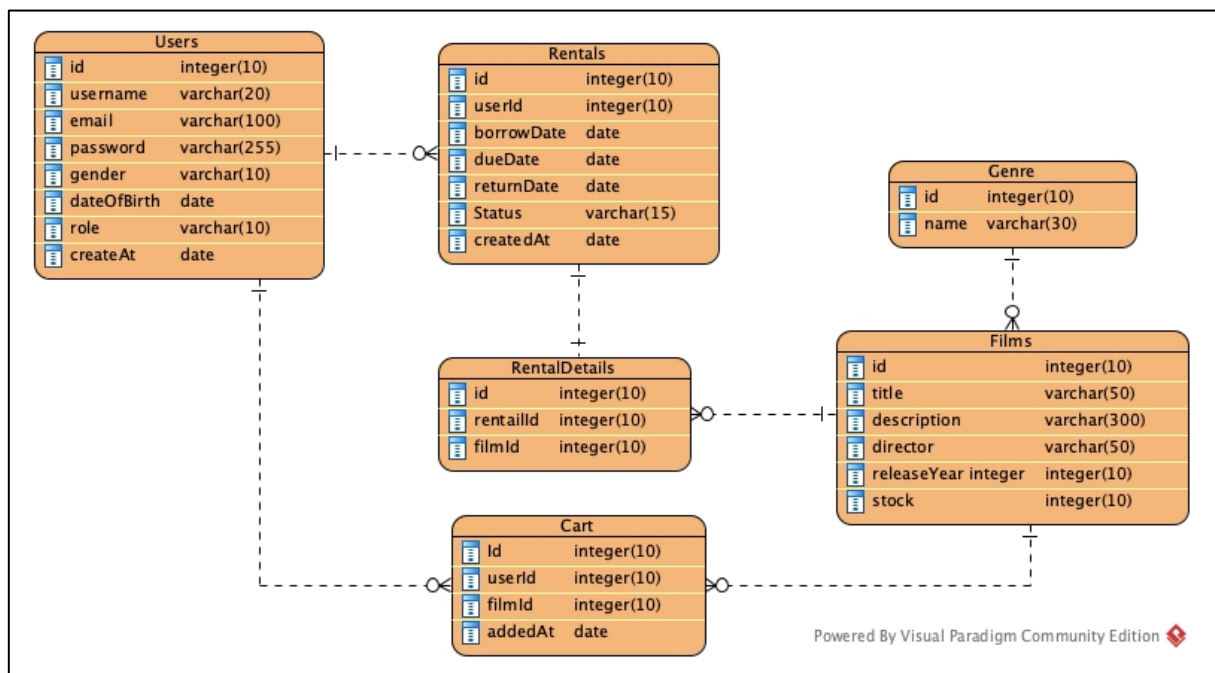


Figure 1. Entity Relationship Diagram

This is the **Entity Relationship Diagram (ERD)** that defines the database structure for the CineRent web. You must **create a SQL query file** to create your database. The ERD consists of six (6) main entities:

- **Users**
- **Films**
- **Genres**
- **Cart**
- **Rentals**
- **RentalDetails**

2. Home

This is the website landing page. You need to build this page based on the following criteria:

- **Navigation Bar**
 - Show **links** to each page based on the **role (Guest, Member, or Admin)**
 - If the **company logo** is clicked, redirect to the **home page**
 - If the **user's name** is clicked, redirect to the **profile page**



Figure 1. Navigation Bar

- **Content**
 - Show the **company name (CineRent)**
 - Show **six (6) random films** taken from the database
 - Each film includes **title, genre**, and a **link** that redirects to the **Film Detail Page**

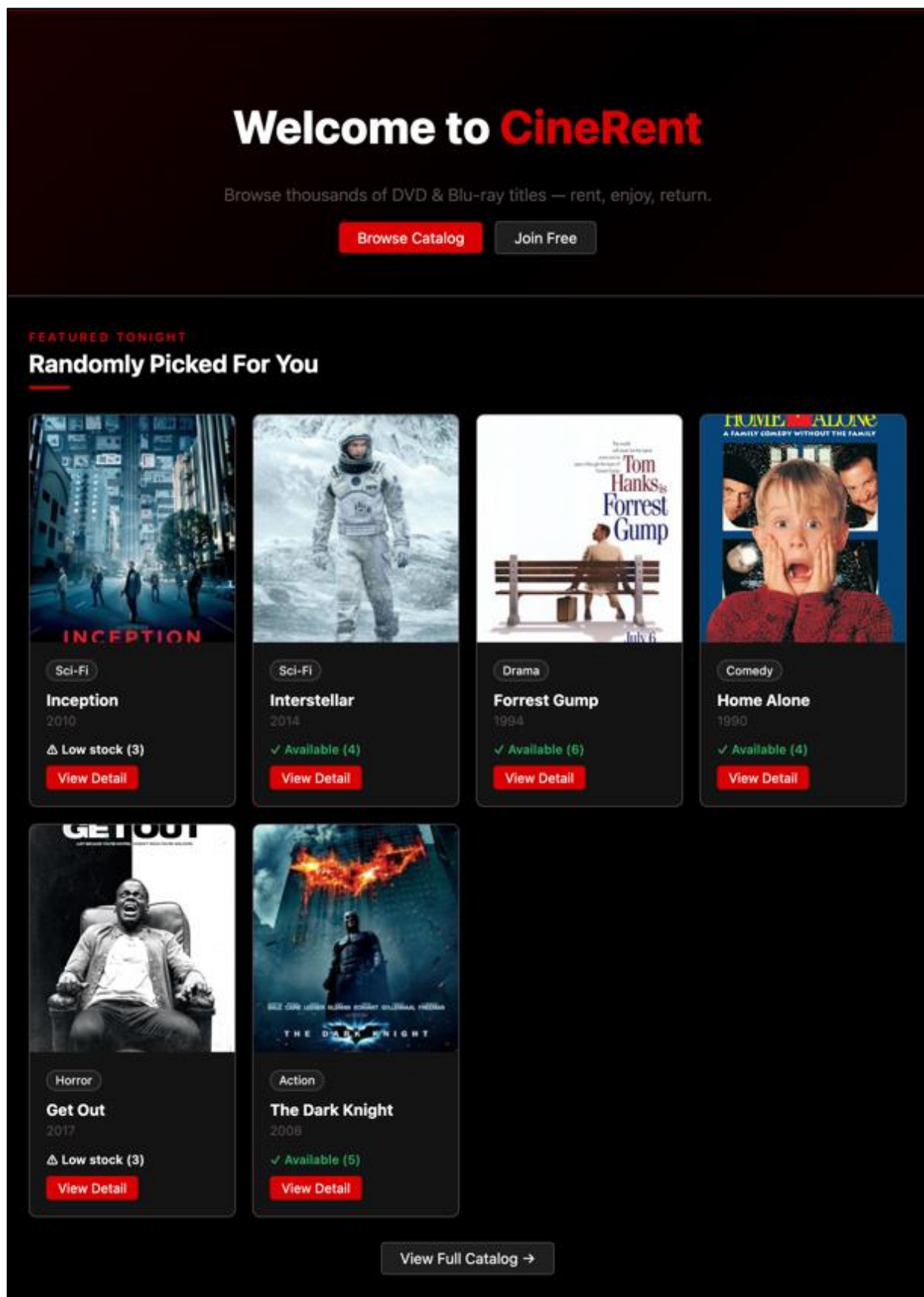


Figure 2. Home Page

- **Footer**

- Show **contact information**. Feel free to design the rest of the footer.

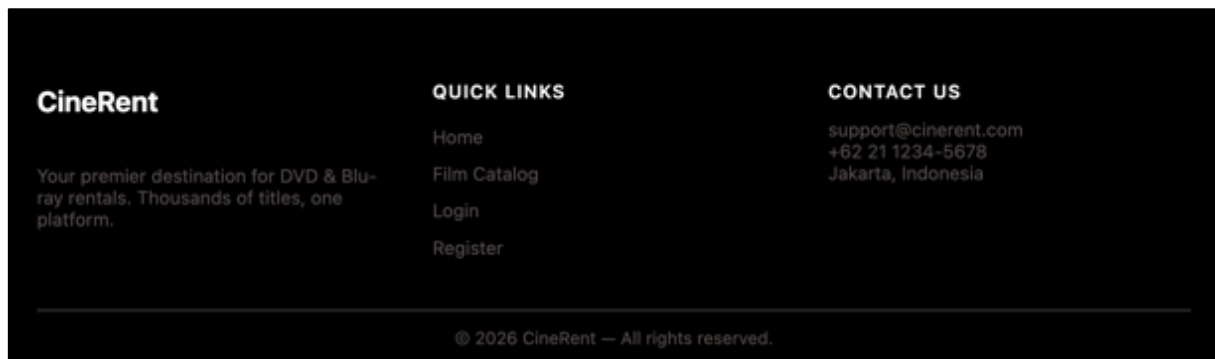


Figure 3. Footer

3. Login

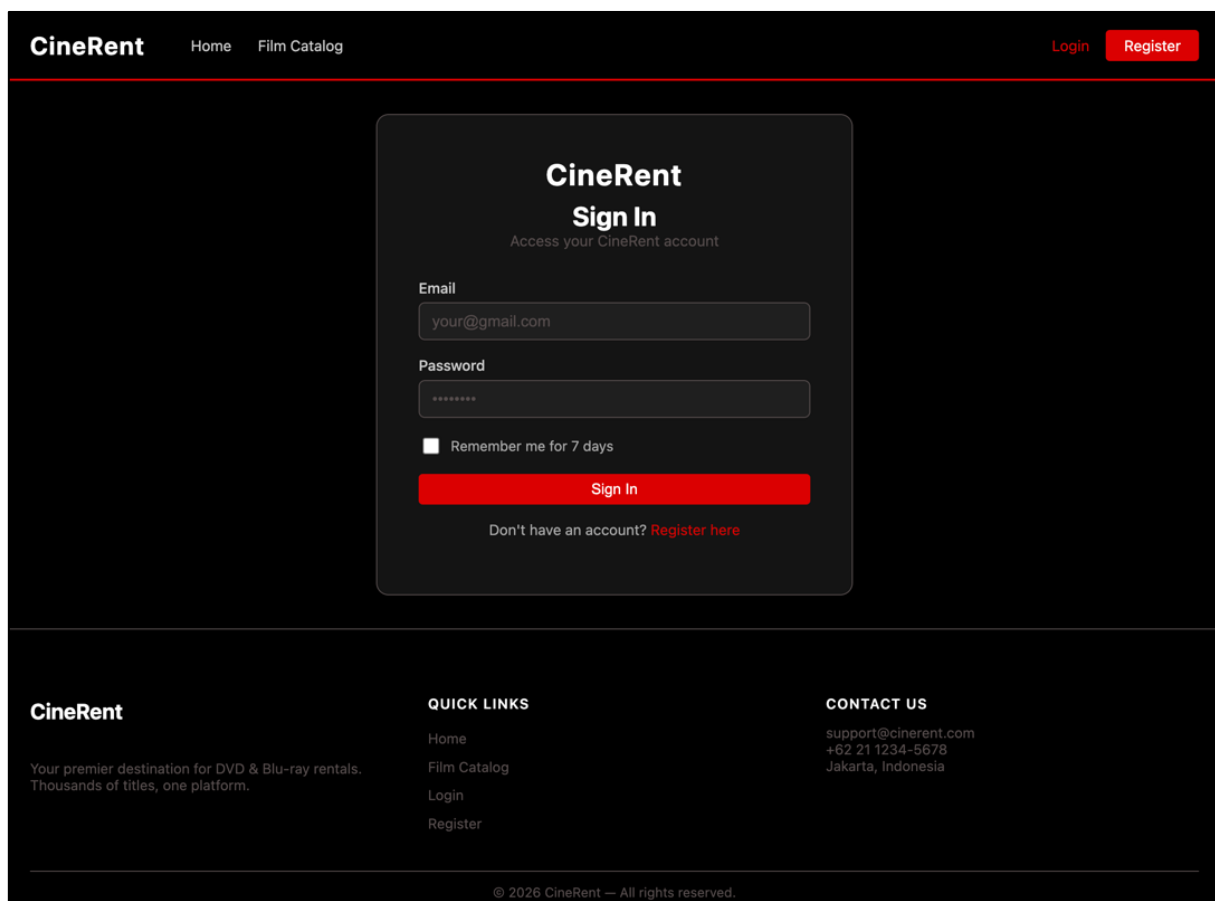


Figure 4. Login Page

This page allows guests to **login** to the website. Display an **error message** on the login page if the guest enters the **wrong credentials**. Build this page based on the following criteria:

Field	Criteria
Email	- Must be filled - Must match with the database
Password	- Must be filled - Must match with the database
Remember me	Add cookie to store User ID when guest login successfully, and will last for 1 week

When the guest logs in successfully, redirect to **home page**.

4. Register

CineRent Home Film Catalog Login **Register**

CineRent
Create Account
Join CineRent and start renting today

Username
4–20 characters

Email
example@gmail.com

Password
Min. 8 characters

Confirm Password
Repeat password

Gender
-- Select --

Date of Birth
dd/mm/yyyy

Create Account

Already have an account? [Sign in](#)

CineRent
Your premier destination for DVD & Blu-ray rentals.
Thousands of titles, one platform.

QUICK LINKS
Home
Film Catalog
Login
Register

CONTACT US
support@cinerent.com
+62 21 1234-5678
Jakarta, Indonesia

© 2026 CineRent — All rights reserved.

Figure 5. Register Page

This page allows guests to **register** themselves to the website. Display an **error message** on the register page if **invalid data is entered**. Build this page based on the following criteria:

Field	Criteria
Username	• Must be filled • 4 – 20 characters • Must be unique
Email	• Must end with '@gmail.com' • Must be unique
Password	• Must be filled • Minimum 8 characters
Confirm Password	• Must match with password
Gender	• Must select 'Male' or 'Female'
Date of Birth	• Must be a valid date in the past

When the guest registers successfully, redirect to **catalog page**.

5. Film Catalog

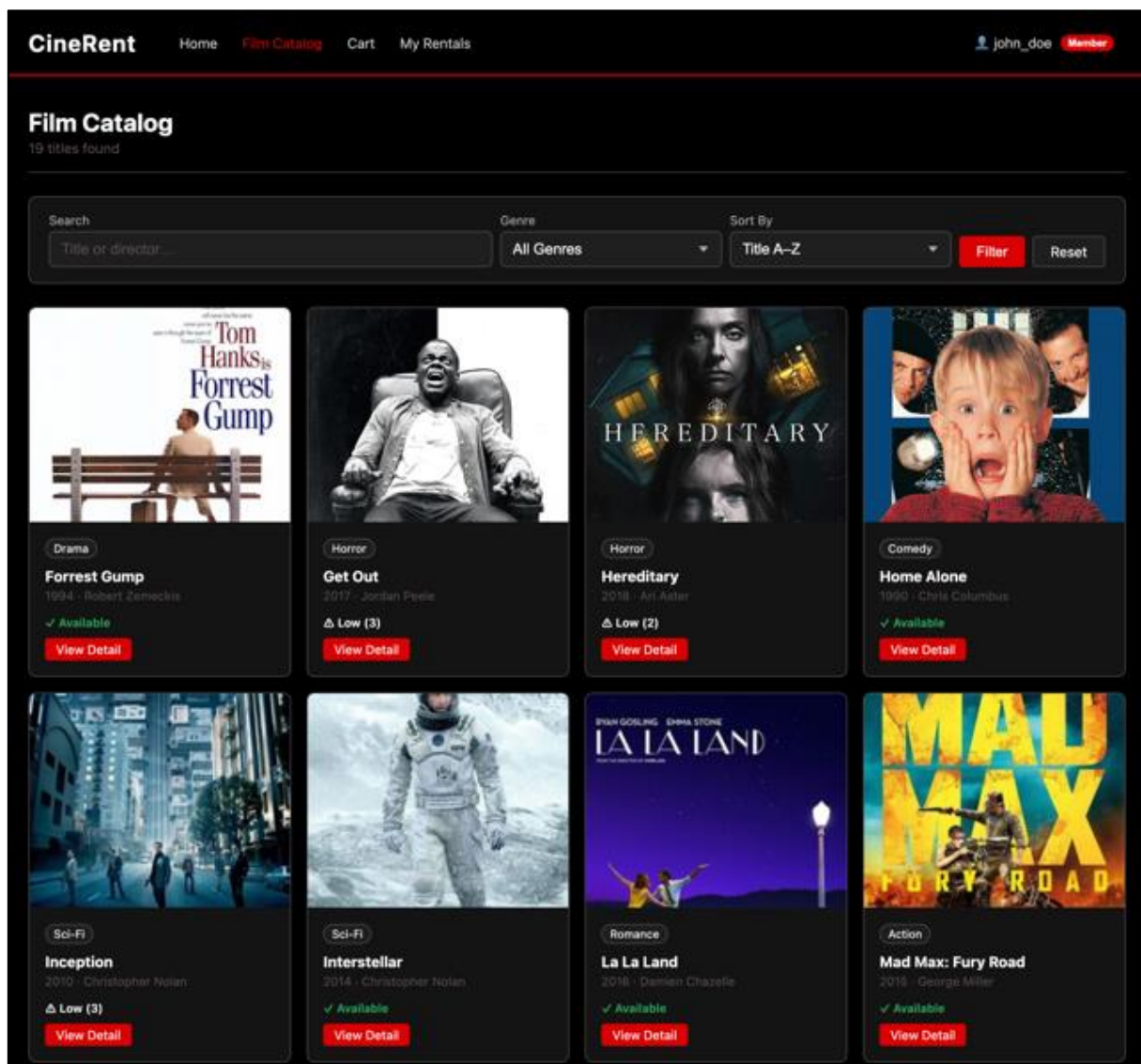


Figure 6. Film Catalog Page

This page allows all roles to **view all available films**. Build this page based on the following criteria:

- Show **film title, director, release year, genre, and description**
- Provide a **"View Detail"** link that will redirect to the film detail page
- Provide **search input, sort, and genre filter**

6. Film Detail

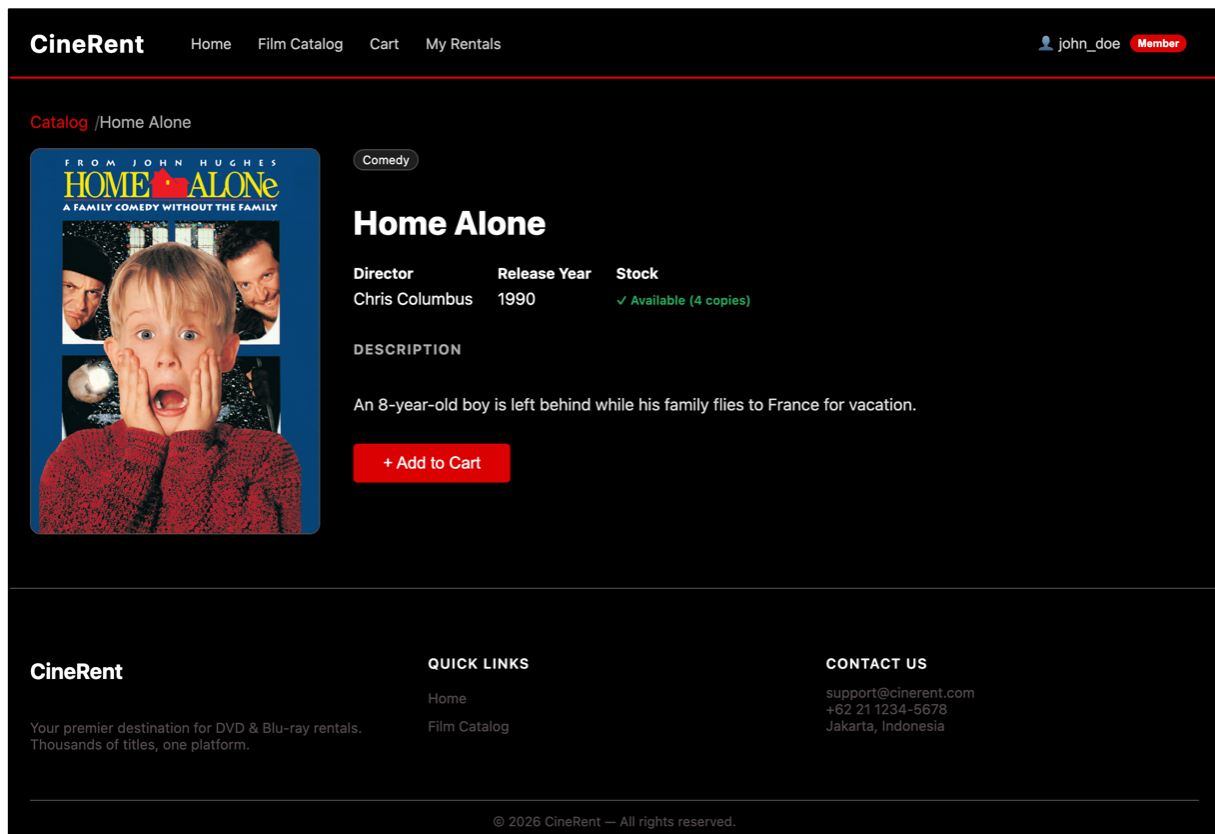


Figure 7. Film Detail Page

This page allows **guests and members** to see **film details**. Build this page based on the following criteria:

- Show **film title, director, release year, genre, description, and stock**
- Show **"Add to Cart" button**
- **Guests** cannot add to the cart and will be redirected to the **login page**

7. Cart

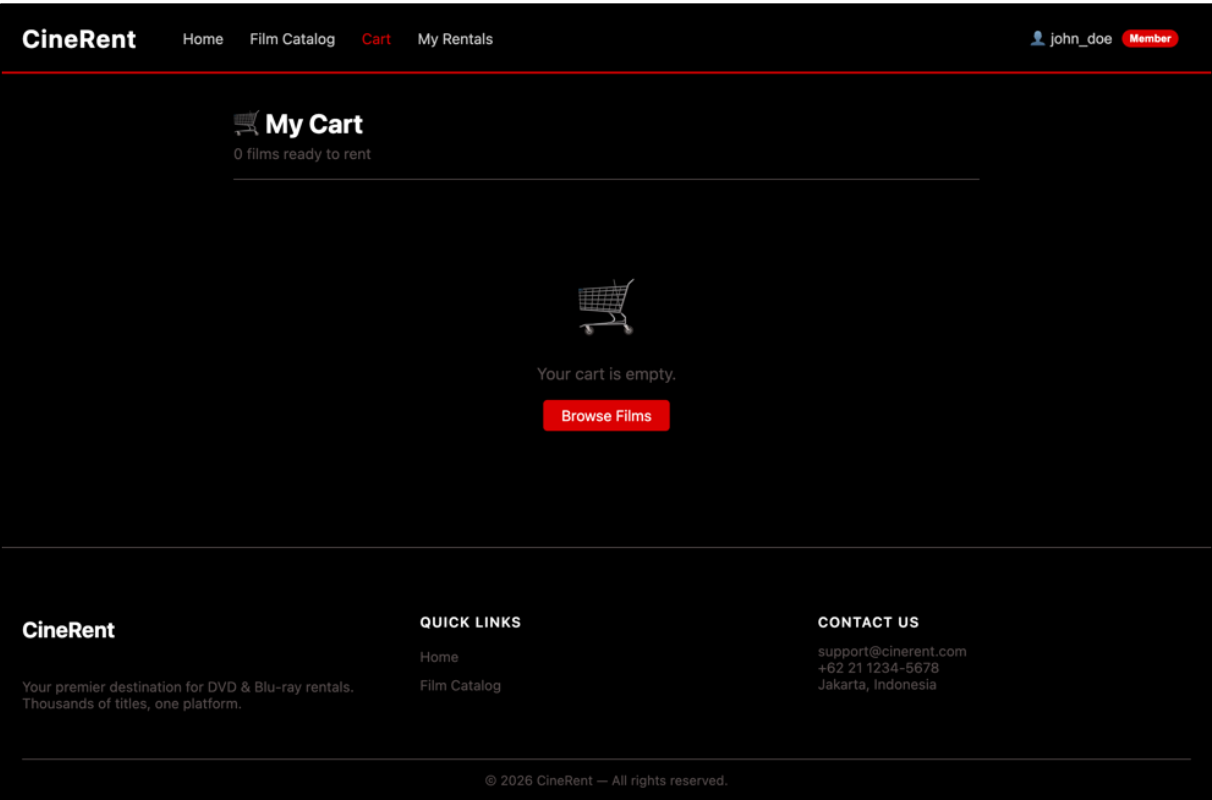


Figure 8. Empty Cart Page

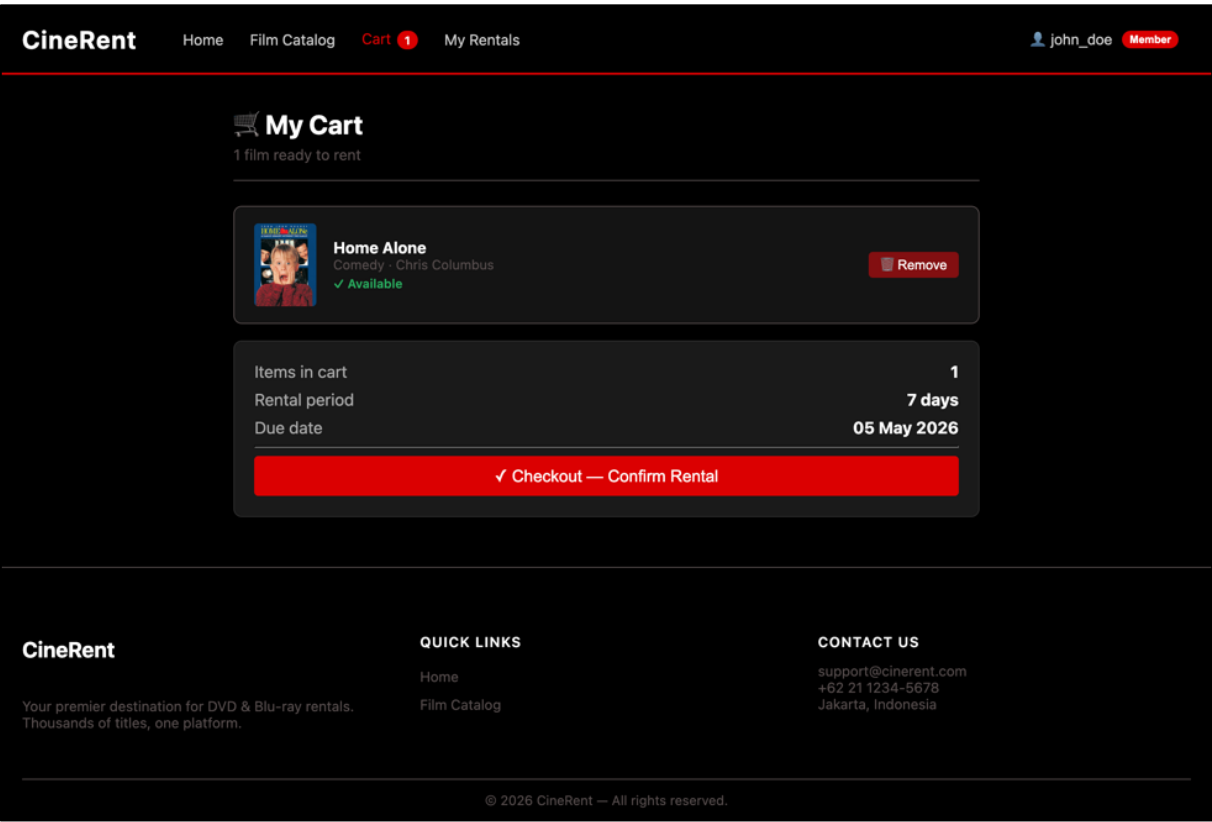


Figure 9. Cart Page

This page allows **members** to **see and manage their cart**. Build this page based on the following criteria:

- Show each **film (title, genre, director)**
- Provide a **delete button** to remove item
- Provide a **"Checkout" button**
- Validate that each film must have **stock greater than 0** to be borrowed

8. My Rentals

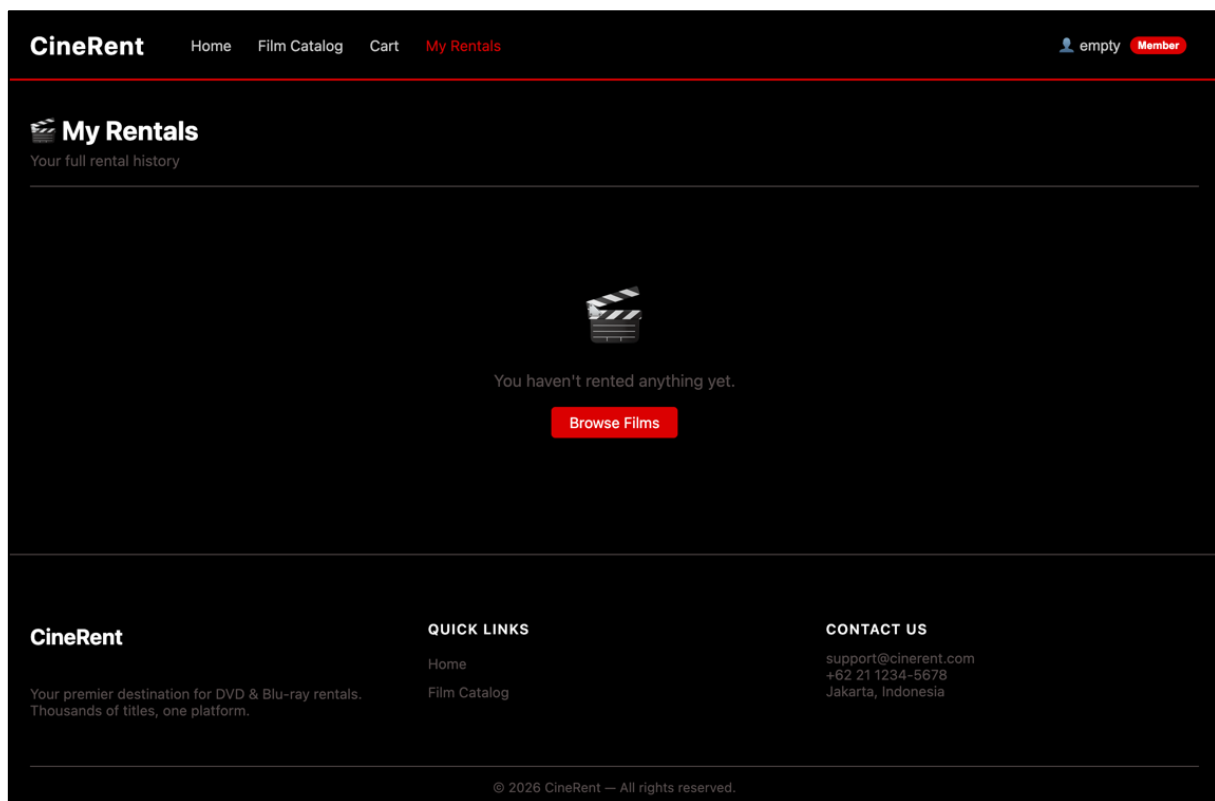


Figure 10. Empty My Rentals Page

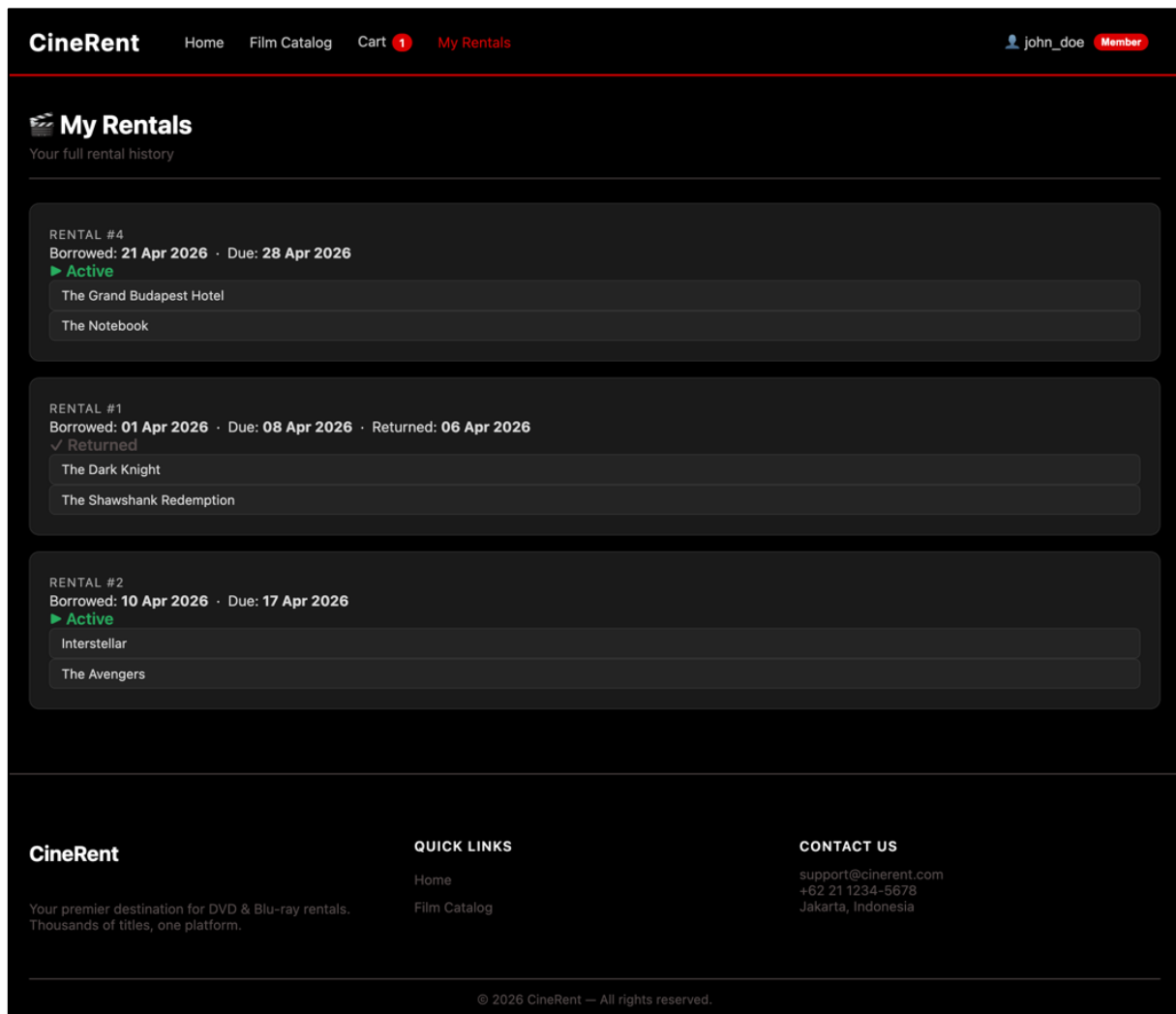


Figure 11. My Rentals Page

This page allows **members** to see their **rental history**. Build this page based on the following criteria:

- Show **all rental records**
- Each rental shows: **rental ID, borrow date, due date, return date, status, and list of films**

9. Profile

Figure 12. Profile Page

This page allows **members and admins** to manage their profile. Build this page with the following sections:

- **Update Profile Info:** Username, Email, Gender, Date of Birth
- **Update Password:** Current password, new password, confirm password
- **Delete Account:** Confirm password, then delete account from database
- **Validate** each field according to register validation

10. Dashboard

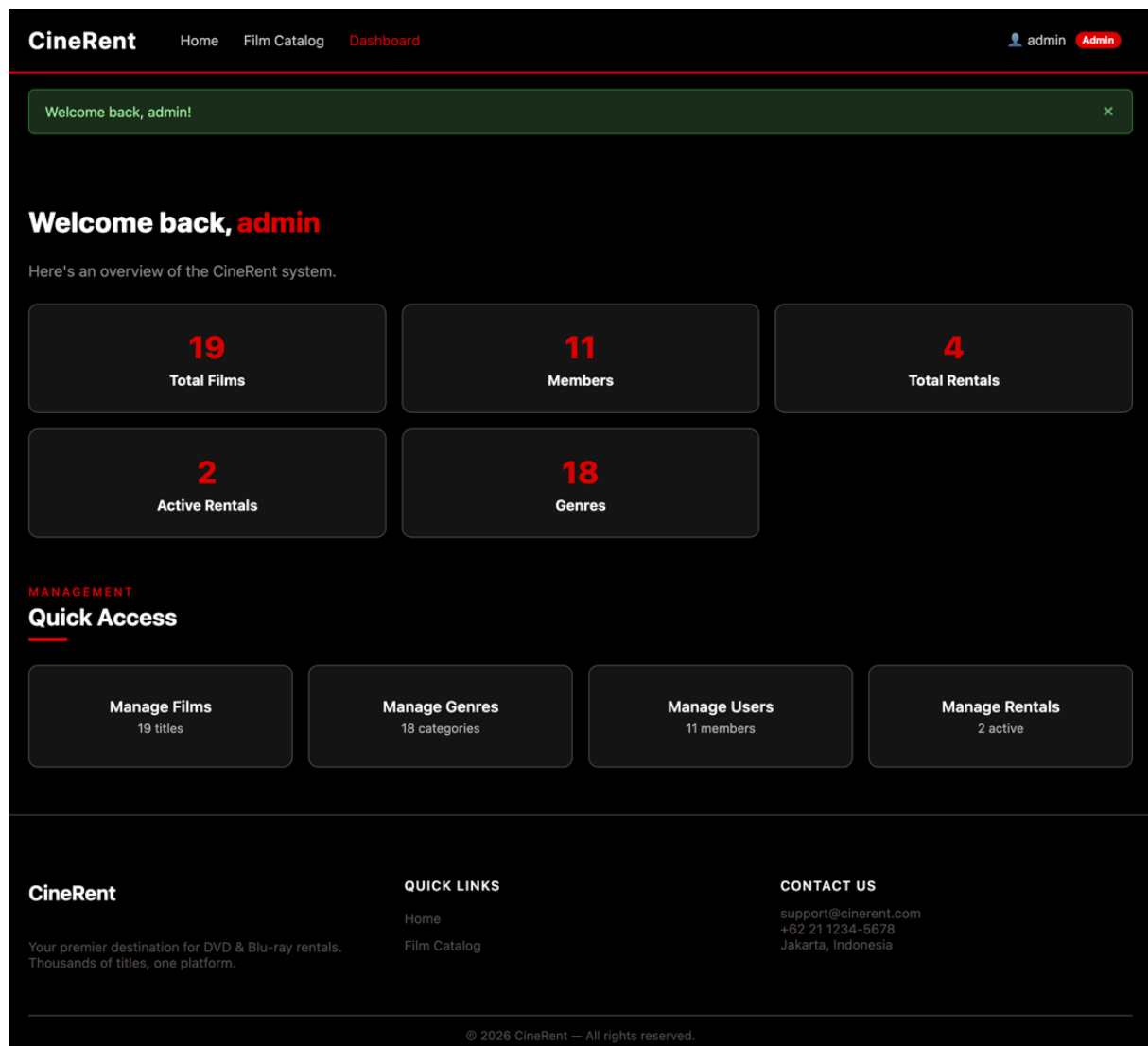


Figure 13. Dashboard Page

This page allows **admin** to view an overview of the system. You need to build this page based on the following criteria:

- Show a **greeting** and **5 navigation cards**:
 - Manage Films
 - Manage Genres
 - Manage Users
 - Manage Rentals
- Each card must be **clickable** and **redirect to the respective page**
- Show **summary statistics**: total films, total members, total rentals, and active rentals

11. Manage Films

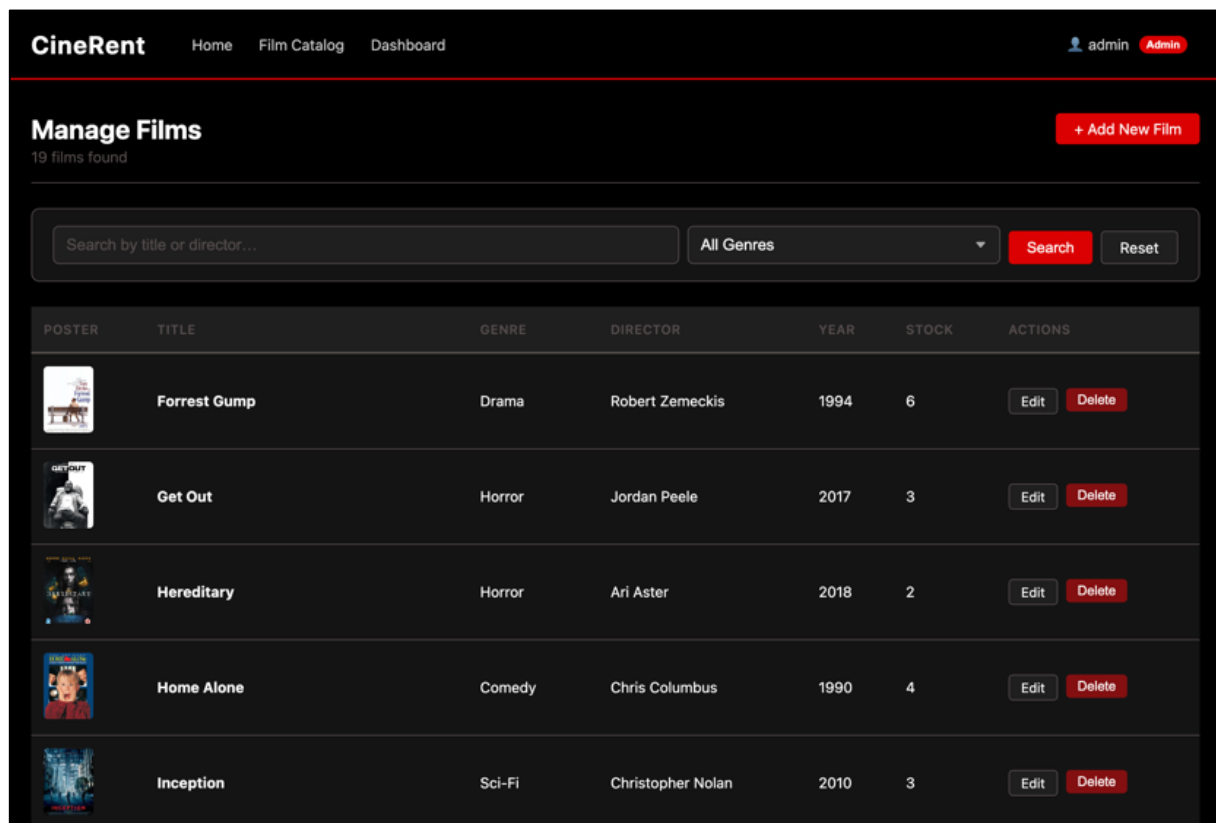


Figure 14. Manage Films Page

This page allows **admin** to manage all film listings. You need to build this page based on the following criteria:

- Show **film ID**, **title**, **genre**, **director**, **release year**, and **image thumbnail**
- Provide an **Add Film** button to redirect to add film page
- Provide a **search input** to search for films based on **title or director**
- Provide a **dropdown filter** to filter films by **genre**
- Provide an **edit button** on each film item that will redirect to the edit film page
- Provide a **delete button** on each film item that will remove the selected film

12. Add and Edit Film

+ Add New Film

[← Back to Manage Films](#)

Film Title *

3–50 characters

Description *

Film synopsis

Director *

Max 50 characters

Release Year *

e.g. 2024

Stock *

0+

Genre *

-- Select Genre --

Poster Image *

Choose File

No file chosen

Accepted: .jpg, .png

+ Add Film

Cancel

Figure 15. Add Film Page

 Edit Film

[← Back to Manage Films](#)

Film Title *

Home Alone

Description *

An 8-year-old boy is left behind while his family flies to France for vacation.

Director *

Chris Columbus

Release Year *

1990

Stock *

4

Genre *

Comedy ▼

Poster Image



Current image

Choose File

No file chosen

Accepted: .jpg, .png (leave blank to keep current)

✓ Save Changes

Cancel

Figure 16. Edit Film Page

This page allows **admin** to **insert new films and update existing film data**. You need to build this form based on the following criteria:

Field	Criteria
Title	• Must be filled • 3 – 50 characters
Description	• Must be filled
Director	• Must be filled • Minimum 50 characters
Release Year	• Must be filled • Must be numeric • Cannot exceed current year
Stock	• Must be filled • Must be numeric • Minimum value is 0
Image	• Must be valid image format (.jpg or .png)
Genre	• Must be chosen from existing genre list

When admin inserts or updates film successfully, redirect to **manage films page**.

13. Manage Genres

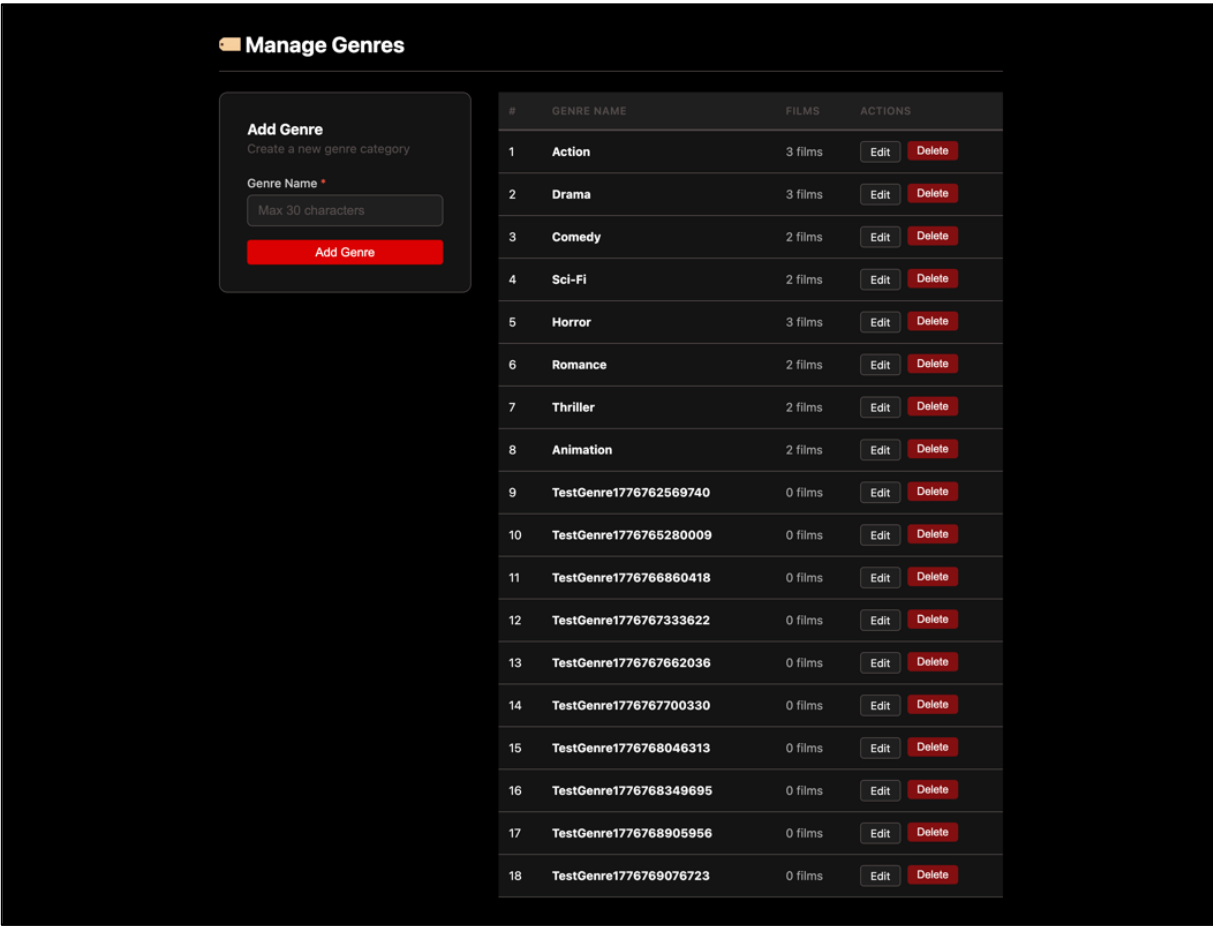


Figure 17. Manage Genres Page

This page allows **admin** to manage all genre data. You need to build this page based on the following criteria:

- Show all genres with their **name** and **film count**
- Provide an **inline Add/Edit Genre form** to add or update genre data
- Provide a **delete button** on each genre item that will remove the selected genre
- Validate that a genre **can only be deleted** if it is **not referenced in any film**

Field	Criteria
Name	• Must be filled • Maximum 30 characters • Must be unique

When admin inserts or updates a genre successfully, redirect to **manage genres page**.

14. Manage Users

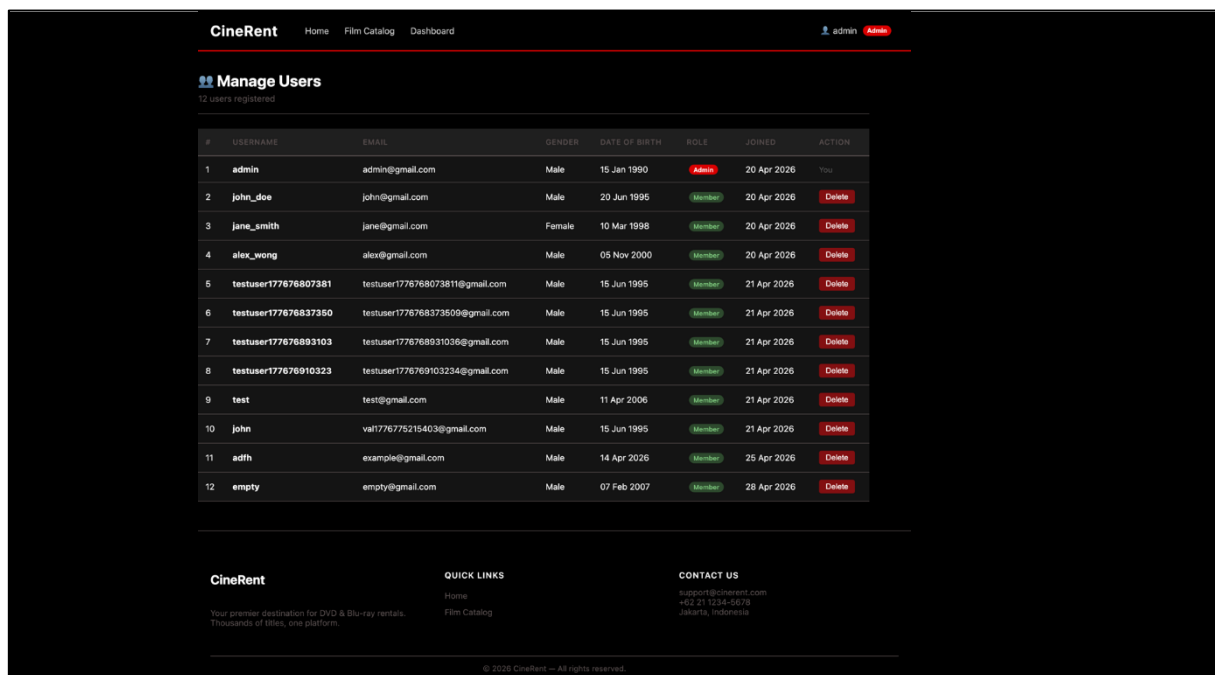


Figure 18. Manage Users Page

This page allows **admin** to **view and delete users**. You need to build this page based on the following criteria:

- Show user's **username, email address, date of birth, gender, and role**
- Provide a **delete button** on each user item that will remove the selected user

15. Manage Rentals

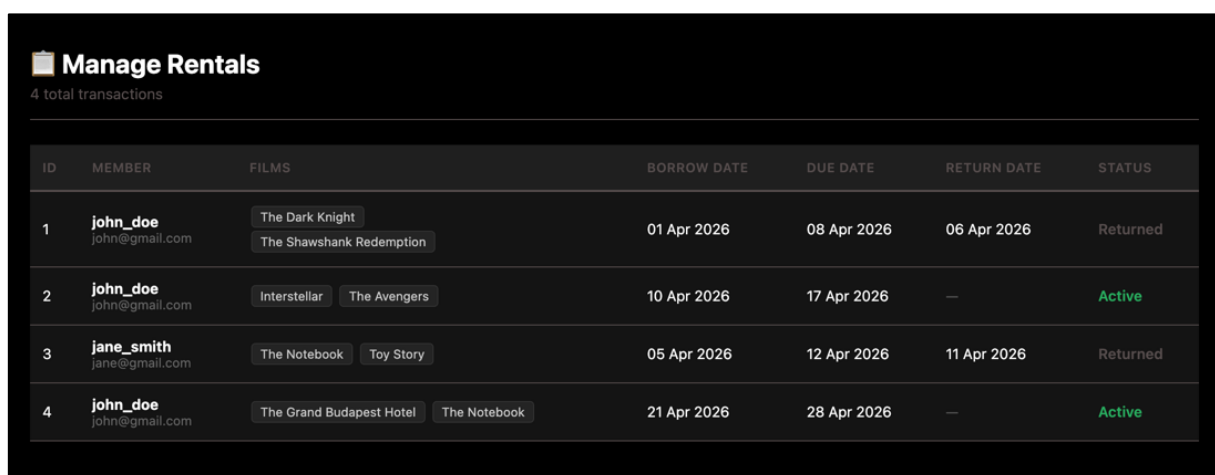


Figure 19. Manage Rentals Page

This page allows **admin** to **view all rental transactions**. You need to build this page based on the following criteria:

- Show **rental ID, username, email, borrow date, due date, return date, status, and list of films**

References:

1. [Forest Gump](#)
2. [Get Out](#)
3. [Hereditary](#)
4. [Home Alone](#)
5. [Inception](#)
6. [Interstellar](#)
7. [La La Land](#)
8. [Mad Max](#)
9. [Parasite](#)
10. [Pulp Fiction](#)
11. [Spirited Away](#)
12. [The Avenger](#)
13. [The Dark Knight](#)
14. [The Budapest Hotel](#)
15. [The Notebook](#)
16. [The Shawshank Redemption](#)
17. [The Silent of the Lambs](#)
18. [Toy Story](#)